

Finding and evaluating community structure in networks

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We propose and study a set of algorithms for discovering community structure in networks—natural divisions of network nodes into densely connected subgroups. Our algorithms all share two definitive features: first, they involve iterative removal of edges from the network to split it into communities, the edges removed being identified using one of a number of possible “betweenness” measures, and second, these measures are, crucially, recalculated after each removal. We also propose a measure for the strength of the community structure found by our algorithms, which gives us an objective metric for choosing the number of communities into which a network should be divided. We demonstrate that our algorithms are highly effective at discovering community structure in both computer-generated and real-world network data, and show how they can be used to shed light on the sometimes dauntingly complex structure of networked systems.